NEW-TO-NATURE CATALYSIS OF A MINIATURIZED HEME-ENZYME BASED ON CARBENE-TRANSFER REACTIONS

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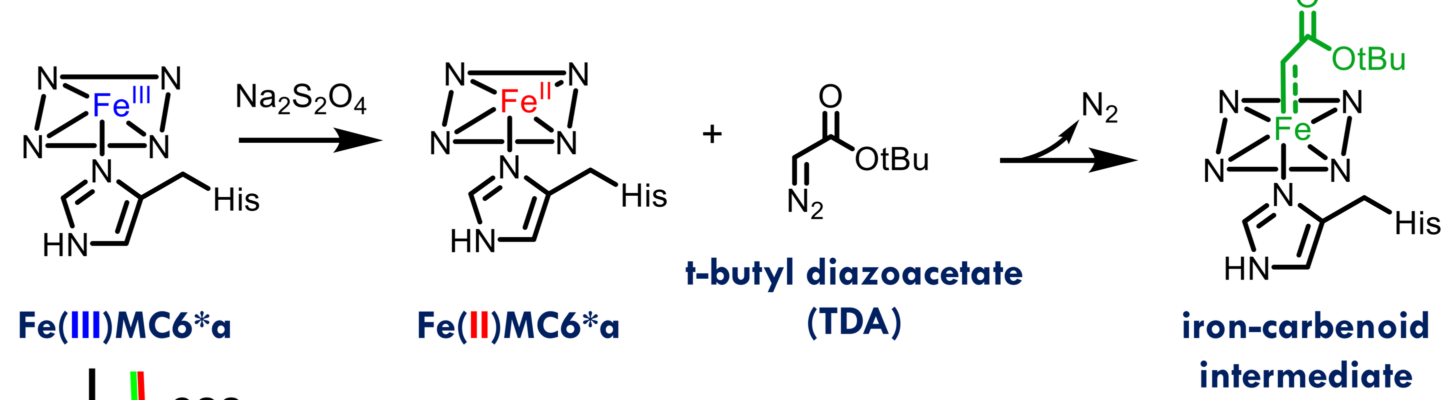
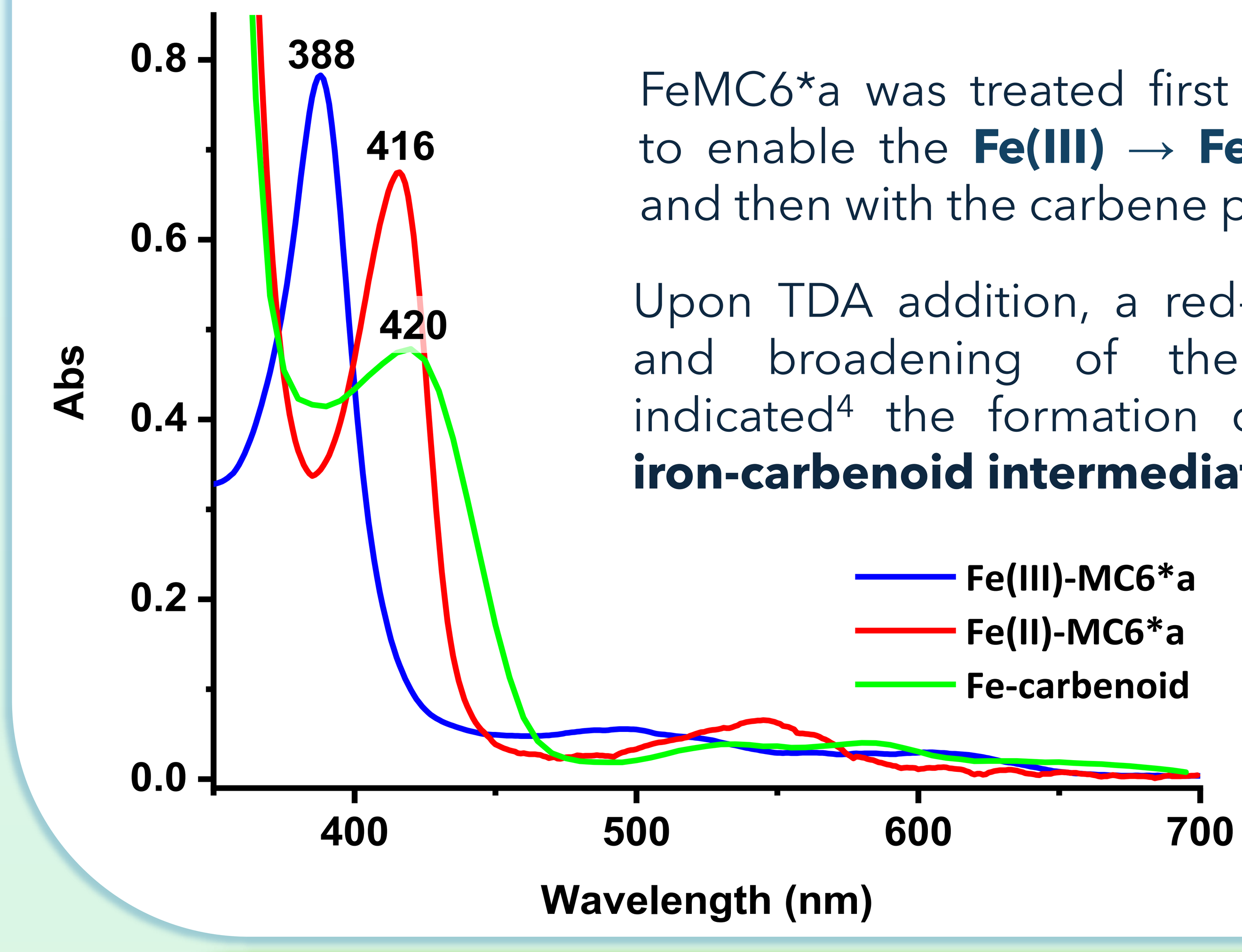
Artificial heme-enzymes (**ArHs**) merge the selectivity of natural biocatalysts with the reactivity of transition metal complexes, enabling **efficient** and **versatile catalysis** beyond natural metalloenzyme functions.[1-3] Recently, the catalytic repertoire of ArHs has expanded beyond biological transformations (typically involving new C-O bonds) to **new-to-nature** reactions,[3] including the formation of C–X bonds (X = C, N, Si, S, B, etc.) through a carbene transfer from the metal ion (a metallo-carbenoid intermediate) to the substrate.[4]

Among ArHs, **Mimochromes** (**MCs**) are **miniaturized**[1,5] heme-enzymes, composed of two short α-helical peptides covalently linked to deuteroporphyrin IX. The latest member of the series, **MC6*a**, stands as an efficient,[6] selective[6,7] and versatile[8] catalyst, acting as a peroxidase,[6] a peroxygenase,[7] a hydrogenase[9] or a CO_2 reductase.[8] The catalytic promiscuity and efficacy of the MC6*a scaffold prompted us to extend the catalytic properties of MCs by investigating their potential to act as a carbene transferase.

UV-Vis characterization: Iron-carbenoid intermediate



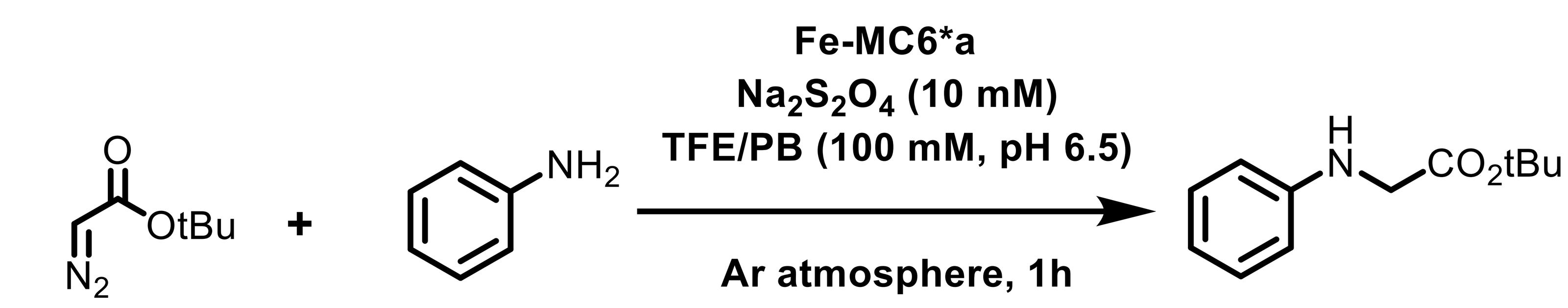


FeMC6*a was treated first with $Na_2S_2O_4$, to enable the **Fe(III) → Fe(II) reduction** and then with the carbene precursor **TDA**

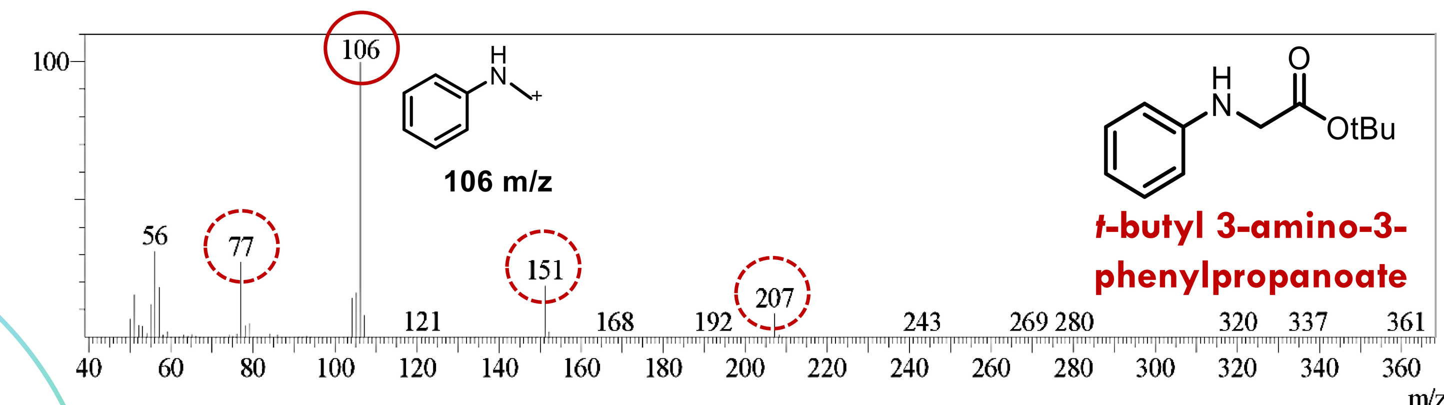
Upon TDA addition, a red-shift, lowering and broadening of the Soret band indicated[4] the formation of a **transient iron-carbenoid intermediate**

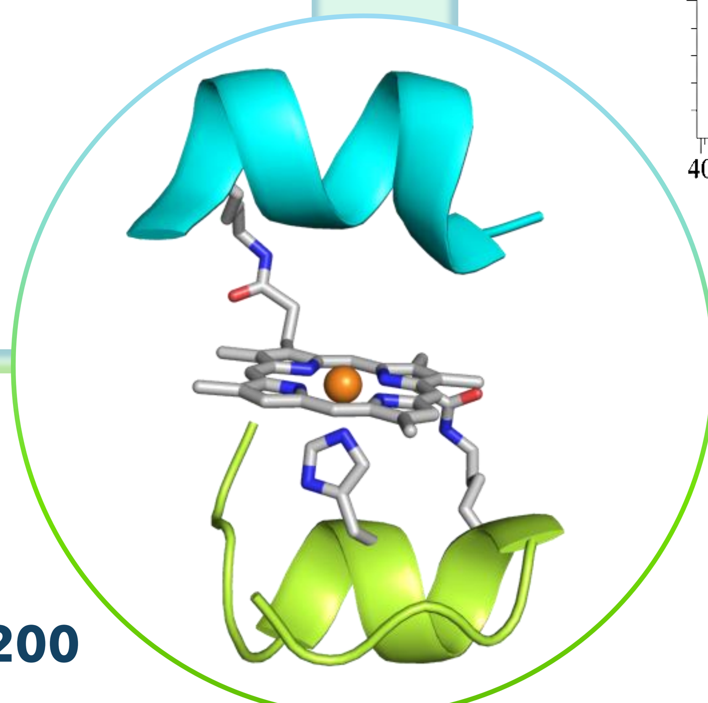
Fe-MC6*a-catalyzed aniline derivatization via N-H insertion

The catalytic potential of FeMC6*a as carbene transferase was tested using aniline derivatization as **N-H insertion** model reaction



GC–MS analysis revealed formation of the **carbene-transfer product** through detection of diagnostic ***N*-methylanilinium radical cation**



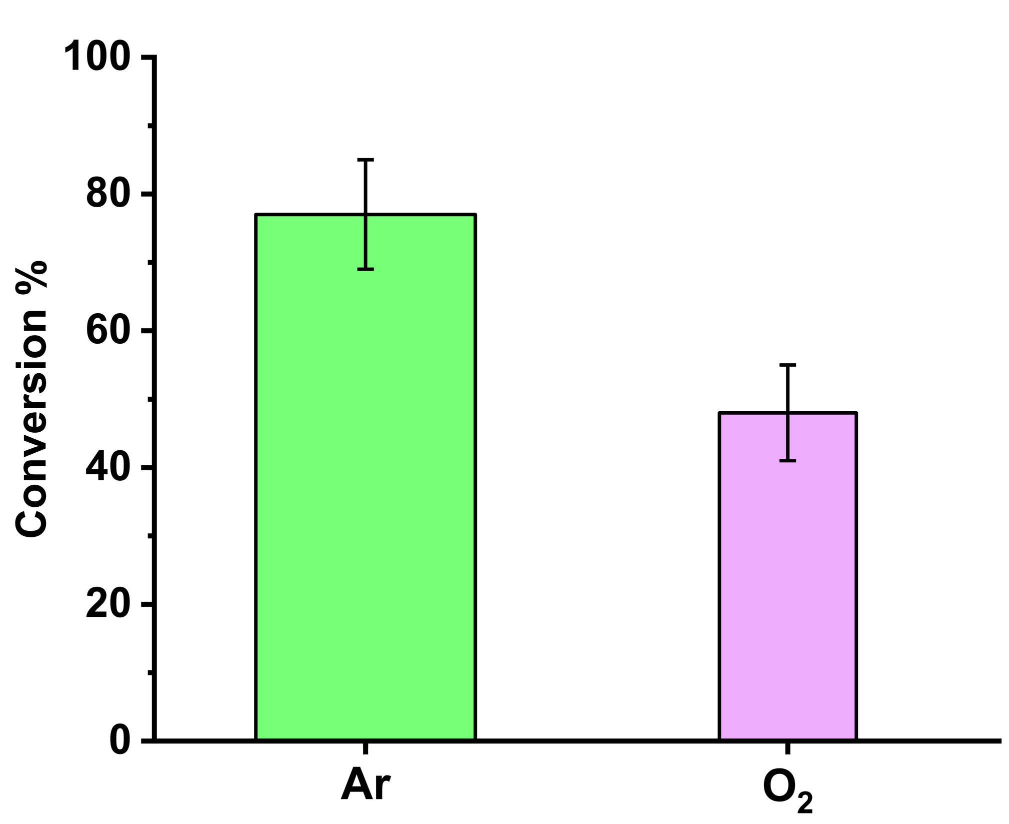


Optimization and impact on O_2 and $Na_2S_2O_3$

The turnover number (**TON**) for this reaction is **2800 ± 200**

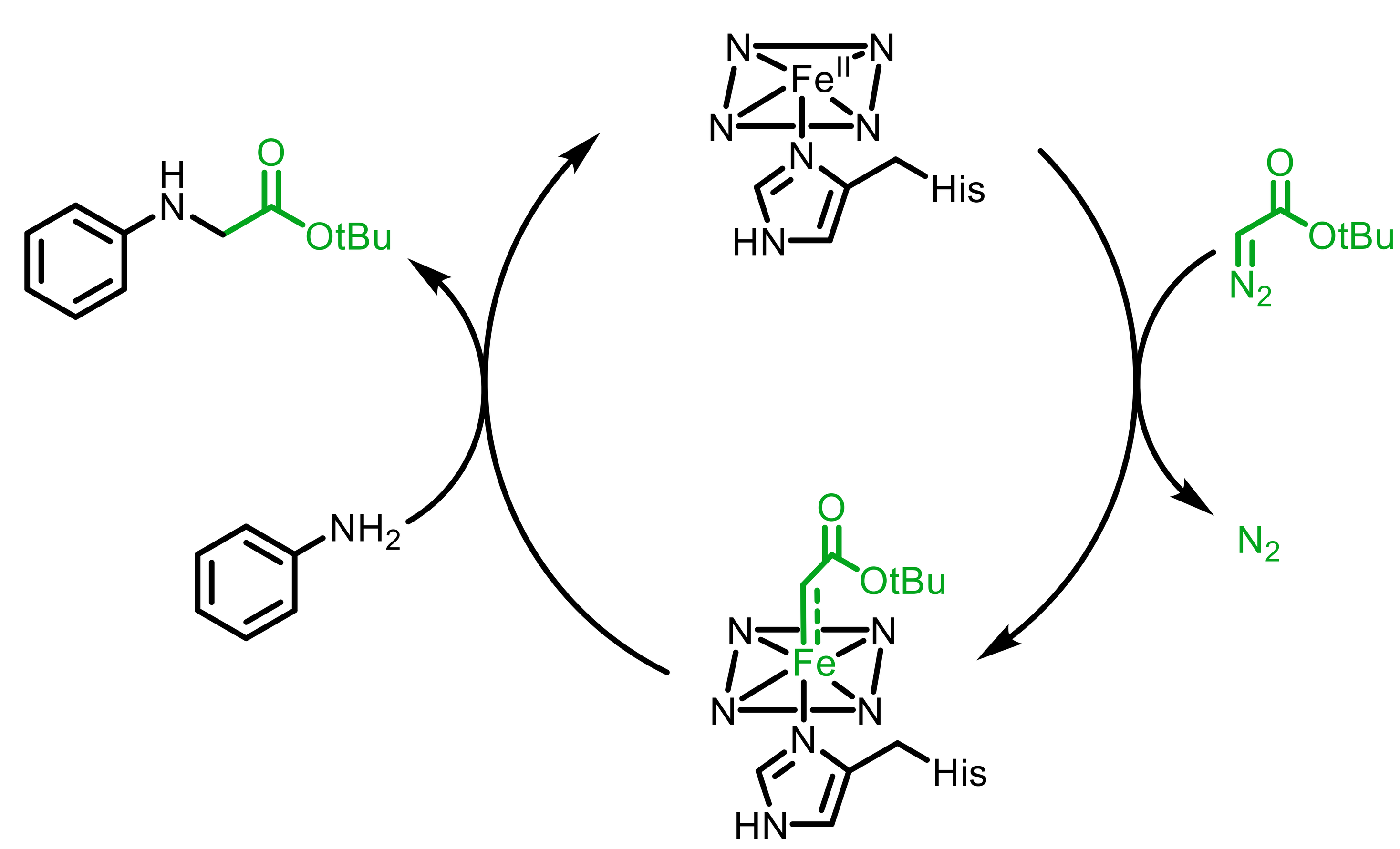
This value surpasses those of **most engineered P450 and Mb variants**[3,10] tested for this reaction

Although the reaction is preferably carried out under **Ar atmosphere,** FeMC6*a retains its catalytic potential even under **O_2 atmosphere**



Mechanistic hypothesis

When **100 eq. of $Na_2S_2O_4$** are used for Fe(III) → Fe(II) reduction, the catalyst still performs **500 turnovers**



This suggests that FeMC6*a cycles between the ironcarbenoid intermediate and a resting **Fe(II) species**

Conclusions and future perspectives

Fe-MC6*a stands as a **promising carbene-transferase**, performing the **abiological N-H insertion reaction** with **thousand turnovers** while working both **under Ar and O_2 atmospheres**. Ongoing work is focused on a comprehensive catalytic evaluation of Fe-MC6*a with a broader range of representative carbene donors.

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